

Integrating Satellite Altimetry and GRACE Gravimetry to Evaluate Reservoir Storage Variations

Key Results

- Peak Surface Storage: Reached a maximum reservoir volume of 88.61 BCM in October 2024.
- 2021 Annual Filling: Added +6.35 BCM/year during initial major impoundment
- 2022 Annual Filling: Accelerated filling added +17.42 BCM/year.
- 2023 Annual Filling: Reached peak annual storage gain of +31.58 BCM/year.
- 2024 Annual Filling: Maintained high impoundment adding +30.15 BCM/year.
- Max Monthly Impoundment Rate: Recorded highest single-month filling in August 2023 at +22.68 BCM/month.
- Seasonal Flux Correlation ($r = 0.61$): Monthly surface change ($\Delta V_{\text{surface}}$) correlates strongly with monthly ΔTWSA during July–September monsoons.
- Multi-Year Trend Correlation ($r = 0.70$): Cumulative surface volume (V_{surface}) strongly aligns with total TWSA anomalies across multi-year filling stages.

Short Explanation

- Monsoon Synchronization: Reservoir impoundment operations rely directly on heavy summer rainy seasons (July–September) rather than year-round continuous filling.
- Satellite Signal Alignment: Strong correlation ($r \approx 0.61 - 0.70$) proves that altimetry-derived surface volume ($\Delta V_{\text{surface}}$) and GRACE gravity signals (TWSA) capture the exact same hydrological filling events.
- Bank Storage Infiltration: Rapid positive spikes in $\Delta V_{\text{surface}}$ during peak filling months force water laterally into unsaturated rock and soil around the reservoir perimeter.

